Laying down the basics

Information abut the course's structure

6 sections
They include quizzes that check your progress

Downloadable slides
Available in Section 1

Downloadable lexicon
Available in Section 6

Staying in touch
Use the Q&A and follow the Announcements

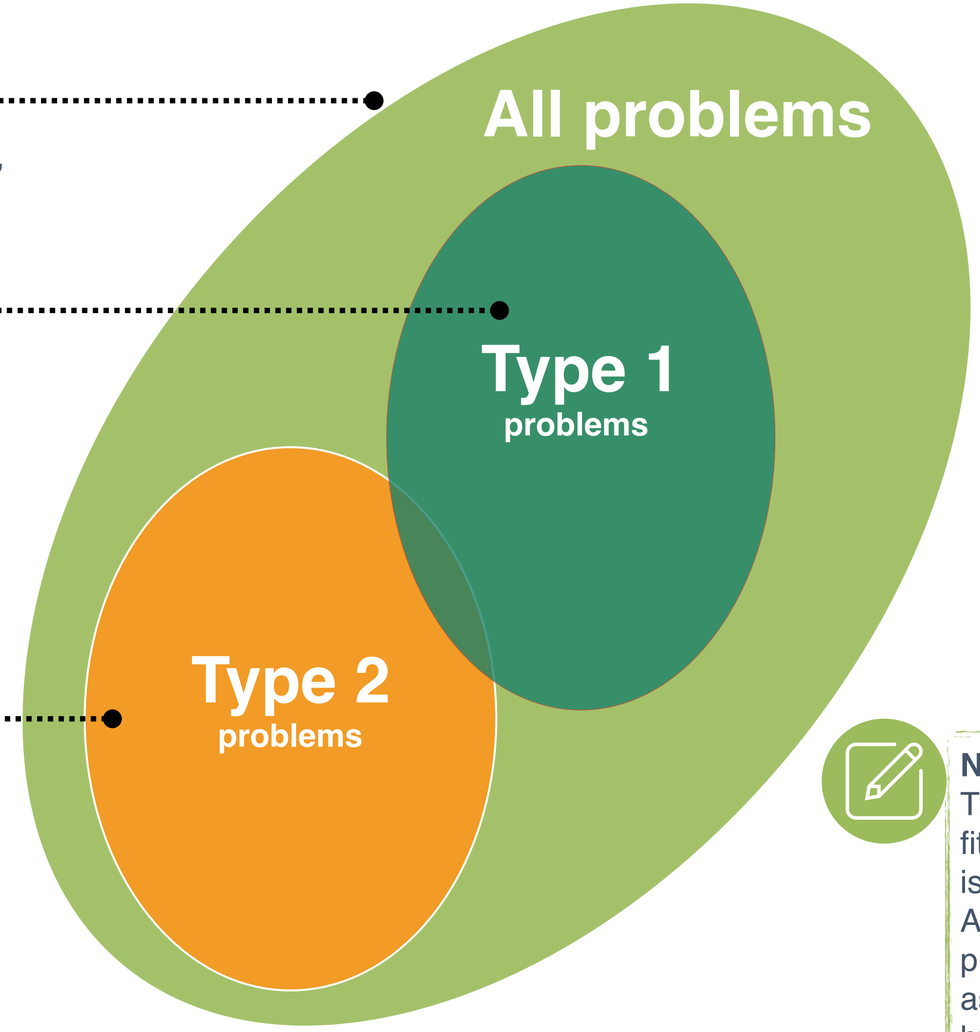
Problems determine solutions

Model of problem categorization

All problems
All kinds of problems encountered, the whole problem space

Type 1 problems
Problems similar to example situations **1** (toothache) and **2** (flat tire). They are solved via **solution-based problem solving approach**, because possible solutions are known from experience. Type 1 problems need **Type 1 experts**.

Type 2 problems
Problems similar to example situations **3** (election) and **4** (acquisition). They are solved via **hypothesis-based problem solving approach**, because no obvious solutions exist. Type 2 problems need **Type 2 experts**.



Note
There are problems that don't fit to either type, this categorization is not exhaustive.
Also, there are problems that could be solved just as well via solution-based and by hypothesis-based approaches.

Not all Experts are the same

Two types of experts

Note
All experts have knowledge, skills and experience. The specific nature of these determines however, what types of problems they can solve

Type 1 Expert	Type 2 Expert
• Many ready solutions in the area of expertise thanks to broad knowledge, skills and experience • Ability to select the right solution for a particular problem situation • Suitable for many every day scenarios and disciplines where problems are repetitive	• Experts at proposing and testing probable solutions to problems while having limited domain knowledge • Ability to isolate the problem's drivers where no obvious solutions are available • Suitable especially to complex business problems that are not well defined

When and why use HBPS?

Applicability and pros of hypothesis-based problem solving (HBPS)

When?

When the problems…

- are not well defined
- are not repetitive
- are not accompanied by a list of solutions to try out
- have a unique setting, circumstances or particular characteristics

When some of the above are true for a given problem HBPS should definitely be considered

Why?

Because when dealing with **not well defined, not repetitive and unique problems that have no obvious solutions**…

- it is faster than the standard appoach of data collection -> analysis -> conclusion
- it requires less effort than the standard approach as only the most promising potential solutions are tested
- it engages all stakeholders of the problem

And it's the method of choice of top consulting firms!

HBPS consists of 4 steps

An iterative approach to problem solving

1 Problem formulation

Know the problem's context and make it specific, measurable, action-oriented, relevant and time-bound (SMART)

2 Problem structuring

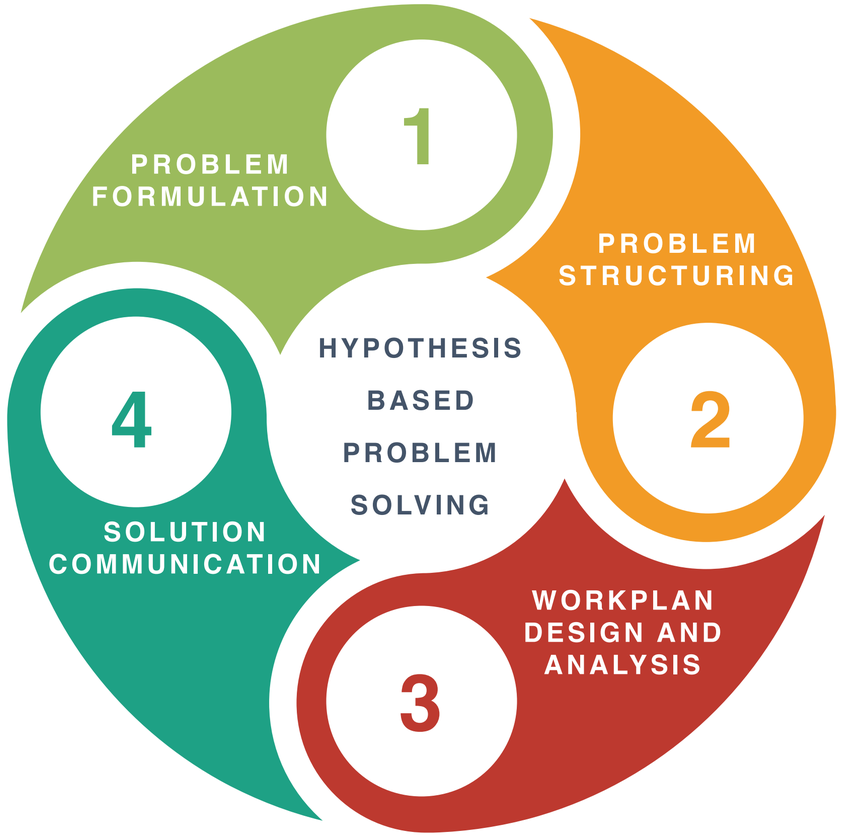
Logically break down the problem into a cascade of manageable sub-problems that cover the whole scope without overlaps

3 Workplan design and analysis

Estimate the required resources and impact of the probable solutions, prioritize the work and start with the most important analyses

4 Solution communication

Formulate the final conclusions and communicate them effectively to the problem's stakeholders



Start with studying the context

Step 1 - Problem formulation

1. **BUSINESS ENVIRONMENT** - The business situation in which the organization finds itself
2. **SUCCESS CRITERIA** - To what extent must the problem be solved to count as a success?
3. **STAKEHOLDERS** - Answer who the decision makers are and who else can influence the engagement
4. **SOLUTION SPACE** - What can be considered as a solution?
5. **SOURCES OF INFORMATION** - Where to look for information relevant to the problem?

Next, formulate the problem right

Step 1 - Problem formulation

ACTION-ORIENTED

Explicitly state the actions that need to be taken to solve the problem

MEASURABLE

Try to quantify the problem as much as possible. It is much easier to work with solid quantifiable objectives rather than qualitatively

RELEVANT

Drop all facts that are not relevant leaving only what is relevant to solving the problem

SPECIFIC

Avoid general problem formulation. Being specific will let you focus the problem-solving

TIME-BOUND

Clearly state what time boundaries apply to the problem and its potential solution

Divide and conquer!

Step 2 - Problem structuring

1

Divide the problem into sensible sub-problems and continue until the sub-sub-… problems you end up with can be tackled by an appropriate analysis. The final **sub-problems cannot overlap**

2

Structuring of the problem **cannot result in a loss** of any pieces of the main problem. When all sub-problems are solved, **in combination they yield the answer** to the main problem.

ME

Mutually Exclusive - No overlaps

Cumulatively Exhaustive - Full problem scope

CE

How to divide?

Step 2 - Problem structuring

"How to grow the revenue of a mobile phone manufacturer by 10% in the next 1 year?”

How to divide?

Step 2 - Problem structuring

"How to grow the revenue of a mobile phone manufacturer by 10% in the next 1 year?"

How to grow the revenue of the consumer segment of mobile phone buyers?

How to grow the revenue of the business segment of mobile phone buyers?

...

Using the segments as criterion of decomposition

How to divide?

Step 2 - Problem structuring

"How to grow the revenue of a mobile phone manufacturer by 10% in the next 1 year?"

How to grow the revenue generated by the public sector sales unit?

How to grow the revenue generated by the VIP sales unit?

How to grow the revenue generated by Small&Medium Enterprises unit?

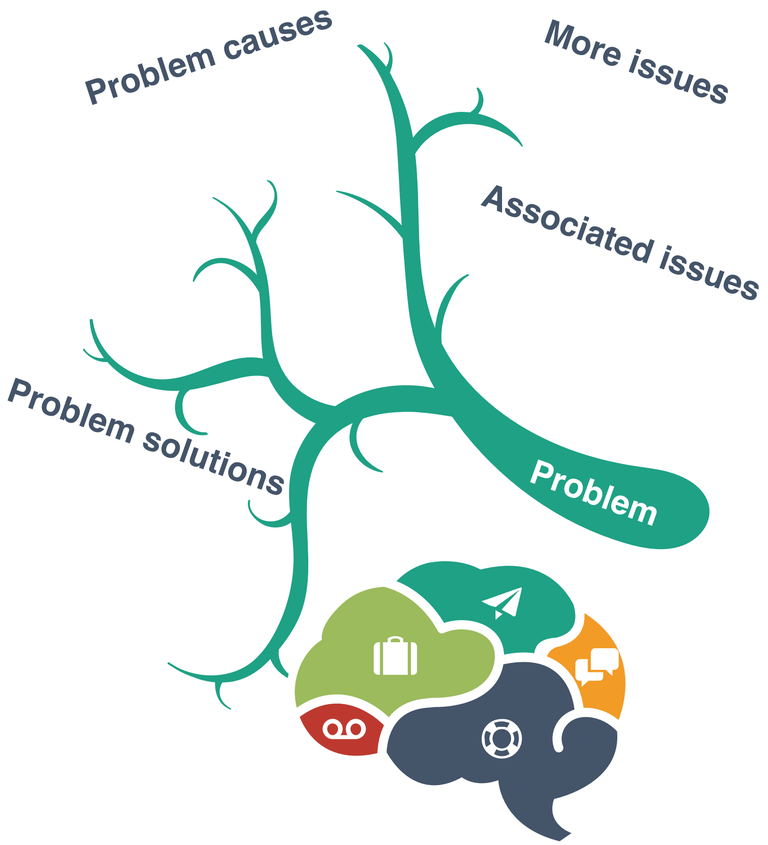
…

Using the organizational unit as criterion of decomposition…

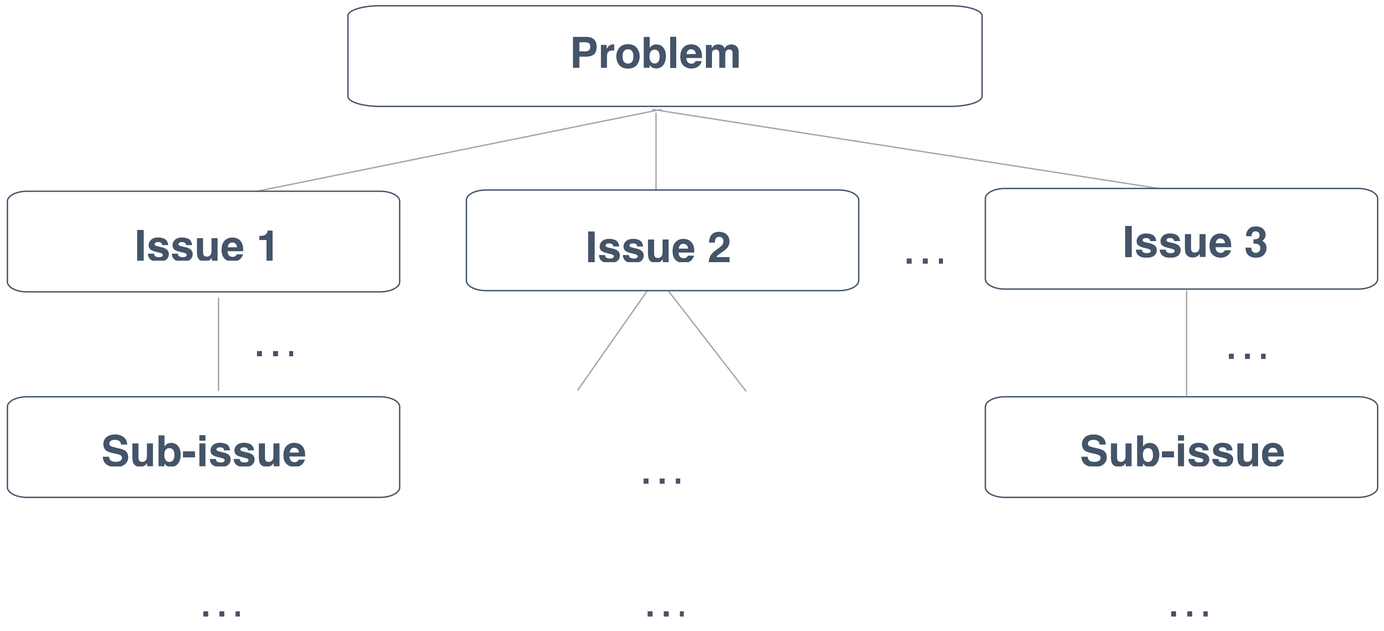
There are always many options of problem decomposition
Efficiency of the subsequent problem analysis is the ultimate measure of how good the decomposition was in the first place. With experience accumulated with successfully solved problems you will decompose optimally right away. If the problem's analysis turns out to be too challenging with a given decomposition, do not refrain from taking a step back and decomposing once more in a different way.

Structuring tools - mind maps and issue trees

Step 2 - Problem structuring



Mind maps are groupings of ideas associated with the central problem. The structure is not rigid and is more of a snapshot of what is related to the problem



Issue trees have a particular structure that ought to be preserved. They depict a cascade of sub-problems (or issues) that together constitute the whole problem. Issue trees are very helpful in tracking the breakdown of a large problem into parts and attempting to solve each part separately

Structuring starts with mindmapping…

Step 2 - Problem structuring

1. Begin with the problem in the centre

Put the representation of the problem in the centre of the drawing space, don't necessarily write but draw and doodle too. Have an open attitude!

2. Branch out your ideas

Put down all ideas (disregarding how crazy they might seem, or who puts them forward) around the centrally placed problem. Common ideas can be connected with branches extending from the centre and further sub-branches

3. Keep going until ideas dry out

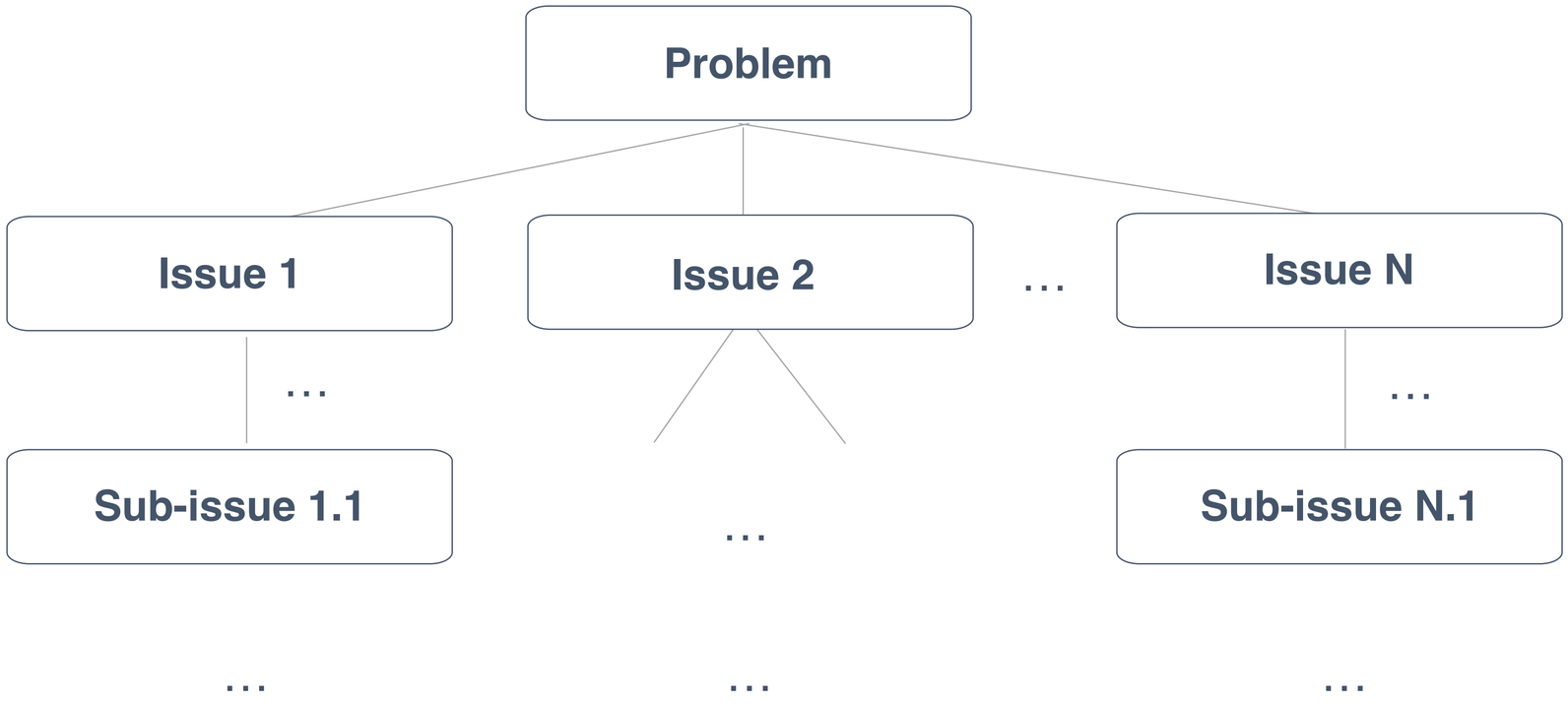
Stimulate the mindmapping exercise to keep the flow of ideas going: use brainstorming kits with colored pens and paper, take turns at the writing board and make breaks. But don't filter the ideas just yet! You will do that when mapping the ideas onto an issue tree

Alternatively use the issue tree right away

Step 2 - Problem structuring

1. Start with the problem at the top
2. Work down decomposing the problem into issues
3. Don't let the issues on the same tree level overlap

Main steps of structuring an issue-tree



Model of the issue tree. Note that the depending on the problem and the particular decomposition method chosen there can be any number of issues stemming from the problem, sub-issues stemming from issues, etc.

Hypotheses are the core of the structure

Step 2 - Problem structuring

Hypotheses formulation - things to note

1. Keep asking what the “so what?” is for every formulated hypothesis. This checks their relevance to solving the problem
2. When on an engagement or when solving a business case make the hypotheses action oriented, not diagnostic
3. Use knowledge specific to the domain of the problem. You don’t need expert knowledge to formulate hypotheses but it will help you along the way
4. Think out of the box, be creative!

...

Sub-issue 1.1 - What opportunities exist for the energy retailer to decrease the price of renewable energy it procures?

Hypothesis 1 - Manage cost of own renewable energy generation

Hypothesis 2 - Manage cost of renewable energy bought from third parties

...

Hypothesis 1.1 - Manage cost of own renewable energy generation

Hypothesis 1.2 - Manage cost of renewable energy bought from third parties

...

Hypothesis 1.1.1 - Reduce the cost by establishing more efficient maintenance procedures — **Hypothesis A**

Hypothesis 1.1.2 - Reduce the cost by outsourcing administrative processes at plants — **Hypothesis B**

Two hypotheses are formulated by following the thicker branches in the above part of the issue tree:

Hypothesis A: Manage the cost of own renewable energy generation by introducing more efficient maintenance procedures at existing power plants

Hypothesis B: Manage the cost of own renewable energy generation by outsourcing administrative processes at existing power plants

Guidelines for hypothesis formulation

Step 2 - Problem structuring

1 KEEP ASKING "SO WHAT?"

…to make sure that every hypothesis is relevant to the main problem.

2 HYPOTHESIS MUST BE ACTION ORIENTED

…not diagnostic. Solve, don't describe.

3 HAVE BASIC DOMAIN KNOWLEDGE

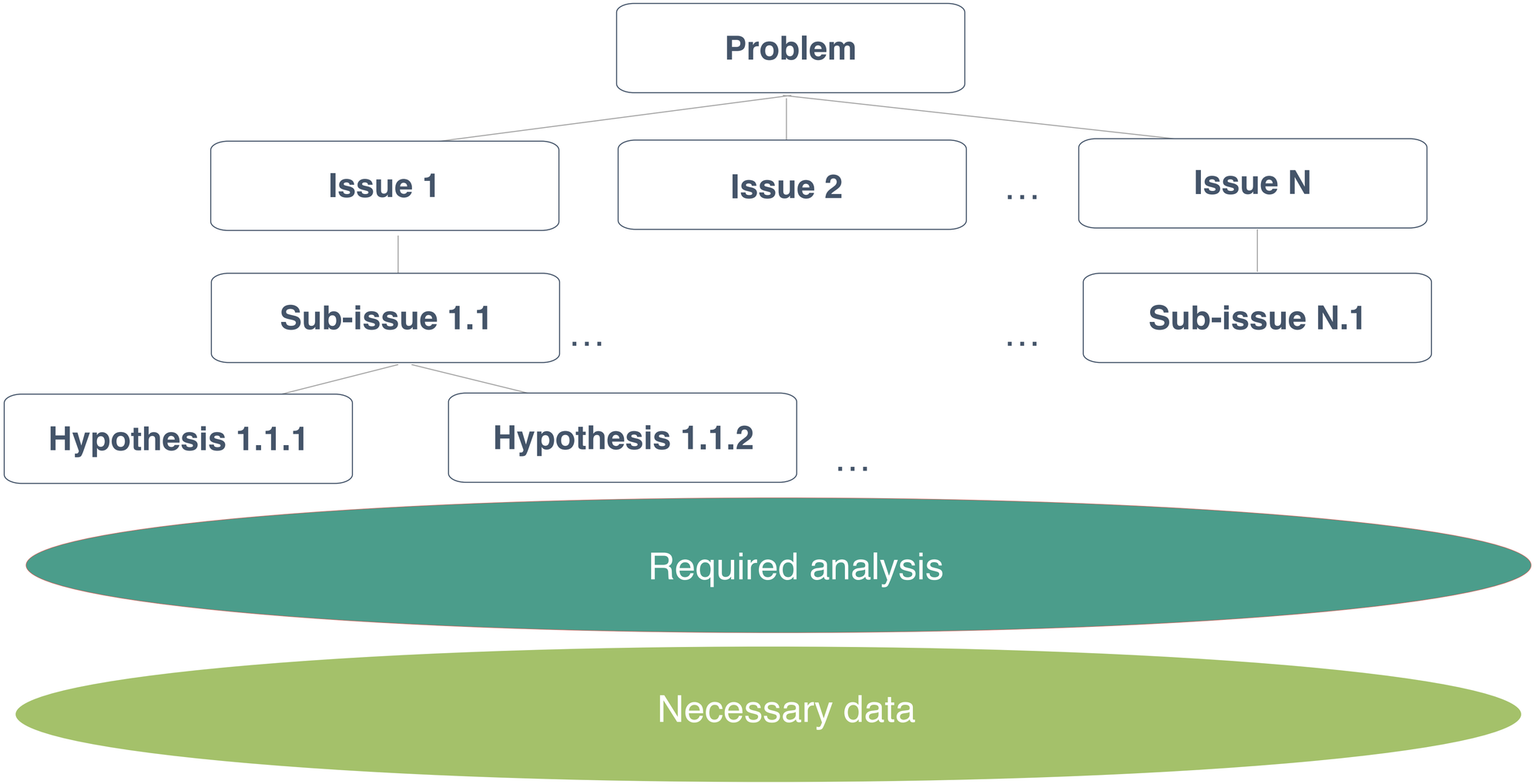
…it will help you in a quicker formulation of more impactful potential solutions

4 BE CREATIVE

…and listen to what everyone has to say

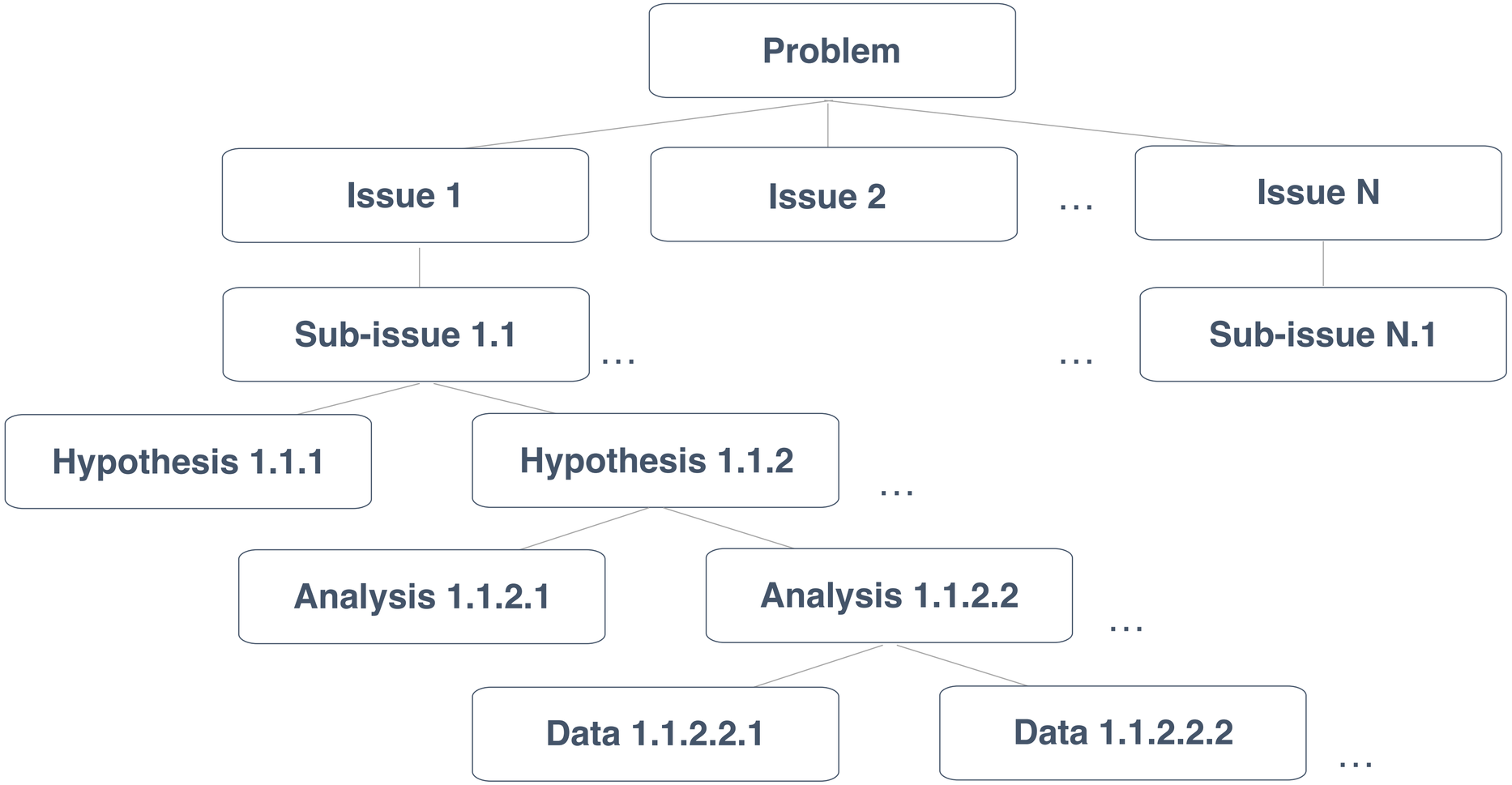
Extending the issue tree

Step 3 - Workplan design and analysis



Extending the issue tree

Step 3 - Workplan design and analysis



Analyses and data validates hypotheses

Step 3 - Workplan design and analysis

Main Problem	Issues	Hypotheses	Analyses	Data
What opportunities exist for the energy retailer to cut costs by 10% by the end of 2018?	What opportunities exist for the energy retailer to decrease the price of renewable energy it procures?	Manage the cost of own renewable energy generation by introducing more efficient maintenance procedures at existing power plants"		
		Manage the cost of own renewable energy generation by outsourcing administrative processes at existing power plants"		
	…	…		
		…		

A neater tabular form of representation of an issue tree when more information (analyses and data) needs to be shown (example problem structure shown)

Example analyses and data

Step 3 - Workplan design and analysis

	Analyses	Data
1	Analysing B2B sales data for 2016	Data regarding sales per sales rep
		Data regarding sales per channel
		Data regarding sales per client
2	Conducting a survey of workforce satisfaction and matching it with salary data	Payroll data
		Input for survey questions
		Survey results
3	Measuring the length of production materials procurement process	Steps of the process
		Average times of steps' completions

Know the reasons and criteria for prioritizing

Step 3 - Workplan design and analysis

Prioritizing work

Reasons:
- Focus work on results
- Help identify quick wins
- Aid the creation of a project plan
- Understand the resource limitations better

Criteria examples:
- Time available for engagement completion
- Human resources
- Know-how resources

Prioritizing hypotheses

Reasons:
- Deliver solutions that fit best the context of the problem

Criteria examples:
- Financial impact
- Ease of implementation
- Time constraint on coming up with a solution, the urgency
- Fit with the organisation's strategy
- Fit with the organisation's culture or values

Start prioritizing right after context study

Step 3 - Workplan design and analysis

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80/20 Rule - the one to remember

Step 3 - Workplan design and analysis

"80 percent of the benefits come from 20 percent of the resources"

First noticed by Vilfredo Pareto in his work *Cours d'Économie Politique* from 1896

- goes by **many names**: 80/20 Rule, Pareto Principle/Law/Theory, etc.
- has **many formulations** that exchange effect and outcomes for outpust/inputs, results/efforts, etc.
- serves as a rule of thumb in **many fields**
- is very often **used in business**
- an extensive description can be found here: http://www.businessballs.com/pareto-principle-80-20-rule.htm

Example from a real life project

Step 3 - Workplan design and analysis

Main problem

"How can intercompany reconciliation be improved by the reduction of time spent on manual reconciliation by 50% in the following 12 months"

Issue 1

Issue 2

…

Issue N

"Implement an IT application at each company of the Group that will specifically aid the reconciliation process"

"Establish a new accounting policy in the Group that will contain better regulation concerning intercompany reconciliation"

"Start an obligatory, regular call concerning reconciliation for all companies of the Group"

Prioritized hypotheses

Prioritization matrix is a well known tool

Step 3 - Workplan design and analysis

Measures can be chosen depending on the specific problem/engagement

Ease of implementation (y-axis) can be measured by:

- time required,
- money required,
- human resources required,
- level of risk involved.

Impacts (x-axis) can be measured by

- percentage amounts of some main goal
- qualitative measures
- industry benchmarks.

Finally, work on prioritized hypotheses starts

Step 3 - Workplan design and analysis

Hypotheses (prioritized)	Analyses	Data	Data sources	Deadlines (for analyses)	Responsibilities
Manage the cost of own renewable energy generation by introducing more efficient maintenance procedures at existing power plants"	Benchmarking of maintenance procedures at power plants	List of maintenance procedures Information on maintenance procedures of other energy retailers	- Generic procedures found in professional literature - List of procedures received from the client company	End of calendar week 9 - internal check	John Doe, Consultant
	Impact measure of new procedures on costs	Data on process costs	- Cost data from industry benchmarks - Cost data from client company	End of calendar week 11 - internal check End of calendar week 12 - planned partial presentation to client	Jane Roe, Senior Consultanr
	…	…	…	…	…

Tabular form of representation of an issue tree for prioritized hypotheses. After prioritization the information regarding data sources, deadlines for analyses and responsibilities can be added in

Explanation starts with synthesis

Step 4 - Communicating the solution

Premises:

1. Prices of **copper** will drop
2. Prices of **silver** will drop
3. Prices of **gold** will drop

Conclusions:

So what?

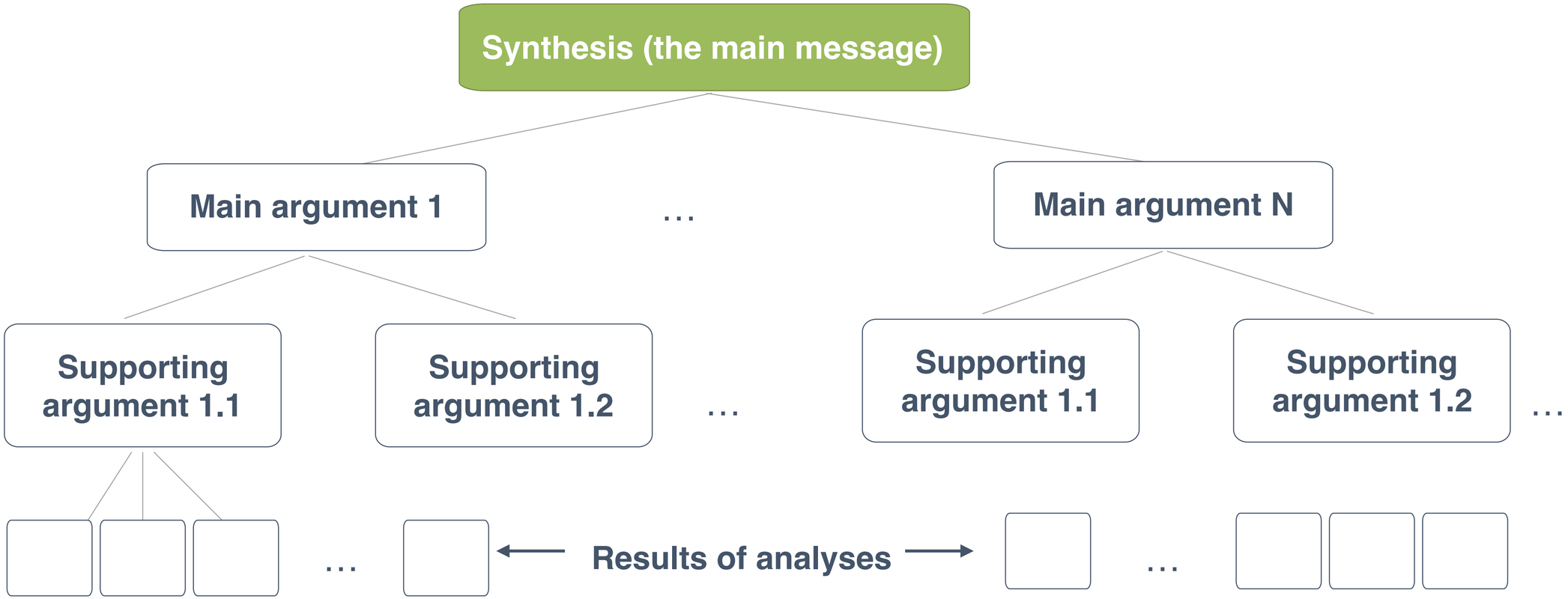
SUMMARY:
“Prices of all three metals will drop”

SYNTHESIS:
"Revenue of the group will tumble if they don't mine more"

Communicate the solution via synthesizing

Step 4 - Communicating the solution

Synthesis is built by asking "so what?"



First communicate the message, then the arguments

Synthesis pyramid and issue tree resemblance
It's not a coincidence that the two look alike. Both follow the logical principle of decomposing aggregate ideas into its constituent parts. This method justifies each piece of communication by referring to actual analyses performed during an engagement.

Pros of synthesizing from day one!

Step 4 - Communicating the solution

1. **Making you stay concentrated on the solution of the main problem**

2. **Strengthening of the logical connections between results of analyses and the final solution**

3. **Reinforcement of prioritization** (as you will track the value added from each analysis from the start)

The story must fit into the context

Step 4 - Communicating the solution

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Importance of context when storyboarding

Every good story will take into account the context of the problem. Especially the first three areas of the context:

- business environment
- success criteria and
- stakeholders

must be accounted for. You have to know what to say (and what not to say!), when to say it and how to say it!

Use storyboarding to prepare the final answer

Step 4 - Communicating the solution

Preparing the slides
The final communication of the solution very often takes the form of a deck of slides. Presenting them is like telling a good story. Good stories have a logical flow and so must your slides and presentation!

1. Outline the story in a sequence of statements

2. Support the statements with relevant data

3. Pay attention to the flow of statements and data

4. Distribute work on the slides between team members

5. Keep the discussion open by accepting different opinion